

Osama Ehab

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SUMMARY

Analytical .NET Developer and Data Science student with a strong foundation in OOP, Clean Architecture, and SQL Server. Passionate about building scalable applications using enterprise-grade patterns and engineering custom code-scaffolding tools to optimize development velocity. Eager to bring algorithmic problem-solving, SOLID principles, and robust system design to a professional software engineering team.

EDUCATION

Alexandria University

B.Sc. in Data Science, Faculty of Computers and Data Science (FCDS)

Alexandria, Egypt

Expected: Jun 2029

PROJECTS

School ERP (MVP) | .NET Developer

Active Development

- **Developing** a modular system utilizing a **Shared Kernel** across a 4-layer **Clean Architecture** (Core, Application, Infrastructure, UI) to centralize core business logic and configurations.
- **Overriding** default **EF Core T4 templates** to intercept the scaffolding process, dynamically enforcing **IEntity/IAuditableEntity** implementations and resolving DB-to-C# naming mismatches.
- **Implementing** **CQS**, **Result Pattern**, and **Generic Repositories** via **Dependency Injection** to enforce strict separation of concerns and reduce boilerplate code.
- **Optimizing** EF Core performance by solving **N+1 query problems**, and applying the **Adapter Pattern** to abstract environment logic (e.g., mapping **ICurrentUser** to a Desktop session).

System Factory: Custom C# Code Scaffolding Engine | .NET Developer

Jan 2026 – Aug 2026

- **Built** an educational **WinForms-based** scaffolding engine leveraging SQL Server **INFORMATION_SCHEMA** to deeply understand the internal mechanics of code generation and ORMs.
- **Developed** a T4-based templating system to generate standard CRUD operations, integrating **Dapper** for high-performance data mapping.

Driving License Management System (DLMS) | .NET Developer

May 2025 – Oct 2025

- **Building** a highly normalized, **database-centric** management system (10+ tables) using **C#**, **WinForms**, and **ADO.NET**.
- **Strictly enforced** a 3-Tier Architecture to securely separate data access, business logic, and presentation layers.

TECHNICAL SKILLS

- **Backend & Frameworks:** C#, .NET, Entity Framework Core (EF Core), RESTful Web APIs, Dapper, ADO.NET
- **Architecture & Design:** Clean Architecture, Shared Kernel, CQS, Repository & Result Patterns, Adapter Pattern, SOLID principles, Dependency Injection
- **Database:** MS SQL Server, Relational Database Design, Metadata-driven Schema Design
- **Tools & UI:** WinForms UI (Data Binding & Configuration)

CERTIFICATIONS

Entity Framework Core | *Programming Advices (Dr. Mohammed Abu-Hadhoud)*

SOLID Principles | *Programming Advices*

Foundations of APIs | *Programming Advices*

Software Engineering Fundamentals Roadmap | *Programming Advices* | *In Progress (20/24 Modules)*